Binary thermodynamic cycles — Solution

X23 (2023) — English translation

A1 0.70 pts

Express the amount of heat Q_- removed from the gas per cycle through T_1, T_2, T_4, R, the amount of gas substance ν and the adiabatic exponent γ.

Solution in TS coordinates.

The figure shows a qualitative diagram of the cycle under consideration in TS coordinates.

Heat is removed from the gas in an isothermal process, so for Q_- we get:

$$Q_- = T_2(S_2 - S_3)$$

where S_2 and S_3 - entropy gas in point 2 and 3 correspondingly. Since change entropy in adiabatic process equal zero:

$$S_2 - S_3 = S_1 - S_4 = \int_{T_4}^{T_1} \frac{\nu C_p dT}{T} = \frac{\gamma \nu R}{\gamma - 1} \ln \frac{T_1}{T_4}$$

from where we find

Solution in pV or pT coordinates.

The diagram shows a qualitative graph of the process in coordinates pV.

We use the Mendeleev-Clapeyron equation:

$$pV = \nu RT$$

Heat is removed from gas in isothermal process, in which $\delta Q = \delta A$, therefore:

$$\delta Q = pdV = \frac{\nu RT dV}{V} \Rightarrow Q_- = \nu RT_2 \int_{V_3}^{V_2} \frac{dV}{V} = \nu RT_2 \ln \frac{V_2}{V_3}$$

Since process 23 isothermal – $V_2/V_3 = p_3/p_2$. Rewrite expression for Q_-:

$$Q_- = \nu RT \ln \frac{p_3}{p_2}$$

For ideal gas in quasi-static adiabatic process correct equation Poisson's:

$$p \sim T^{\gamma/(\gamma-1)}$$

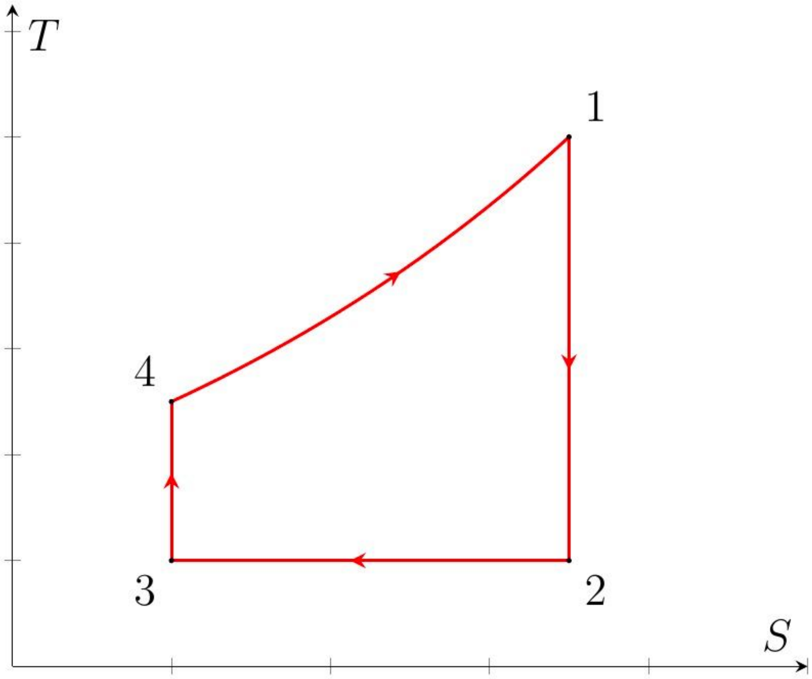
therefore:

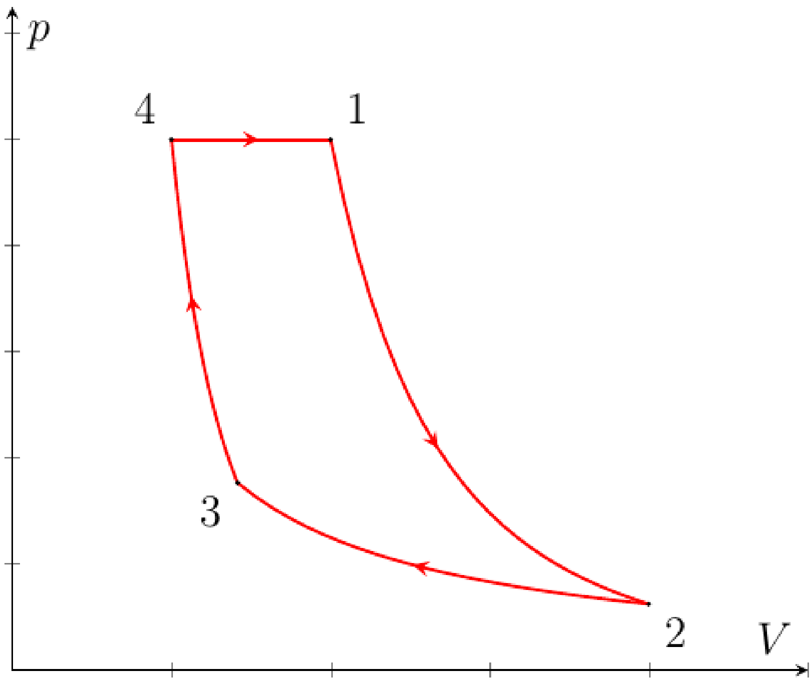
$$\begin{cases} \dfrac{p_1}{p_2} = \left(\dfrac{T_1}{T_2}\right)^{\gamma/(\gamma-1)} \\ \dfrac{p_4}{p_3} = \left(\dfrac{T_4}{T_3}\right)^{\gamma/(\gamma-1)} \end{cases} \Longrightarrow \frac{p_3}{p_2} = \left(\frac{T_1}{T_4}\right)^{\gamma/(\gamma-1)}$$

Since $T_3 = T_2$ and $p_4 = p_1$. Finally:

Answer:

$$Q_- = \frac{\gamma \nu RT_2}{\gamma - 1} \ln \frac{T_1}{T_4}$$





A2 0.80 pts

Express the relation T_1/T_4 in terms of α, β and γ.

Let us write the adiabatic equation for the process $1-2$ in coordinates pT:

$$p^{1-\gamma}T^{\gamma} = const \Rightarrow \frac{T_1}{T_2} = \left(\frac{p_1}{p_2}\right)^{\frac{\gamma-1}{\gamma}}$$

Since $p_1 = p_4$

$$\frac{T_1}{T_2} = \beta^{\frac{\gamma-1}{\gamma}}$$

Also, since $T_2 = T_4/\alpha$:

Answer:

$$\frac{T_1}{T_4} = \frac{\beta^{\frac{\gamma-1}{\gamma}}}{\alpha}$$

A3 — 0.50 pts

Obtain an expression for the cycle efficiency η_1. Express your answer in terms of α, β and γ. Find the numerical value of η_1 for $p_2 = 15$ kPa, $p_4 = 60$ kPa, $T_2 = 340$ K and $T_4 = 400$ K.

Since $Q_+ = C_p(T_1 - T_4)$, we have:

$$\eta_1 = 1 - \frac{C_p T_2 \ln(T_1/T_4)}{C_p(T_1 - T_4)}$$

whence

For α and β we have:

$$\alpha = \frac{20}{17} \qquad \beta = 4$$

from where:

Answer:

$$\eta_1 = 1 - \frac{\ln\left(\beta^{\frac{\gamma-1}{\gamma}}/\alpha\right)}{\beta^{\frac{\gamma-1}{\gamma}} - \alpha}$$

B1 — 0.50 pts

Find the ratio of the masses of mercury and water $m_{(m)}/m_{(w)}$ participating in the cycle.

Solution in TS coordinates.

The figure shows qualitative diagrams of the considered cycles in TS coordinates.

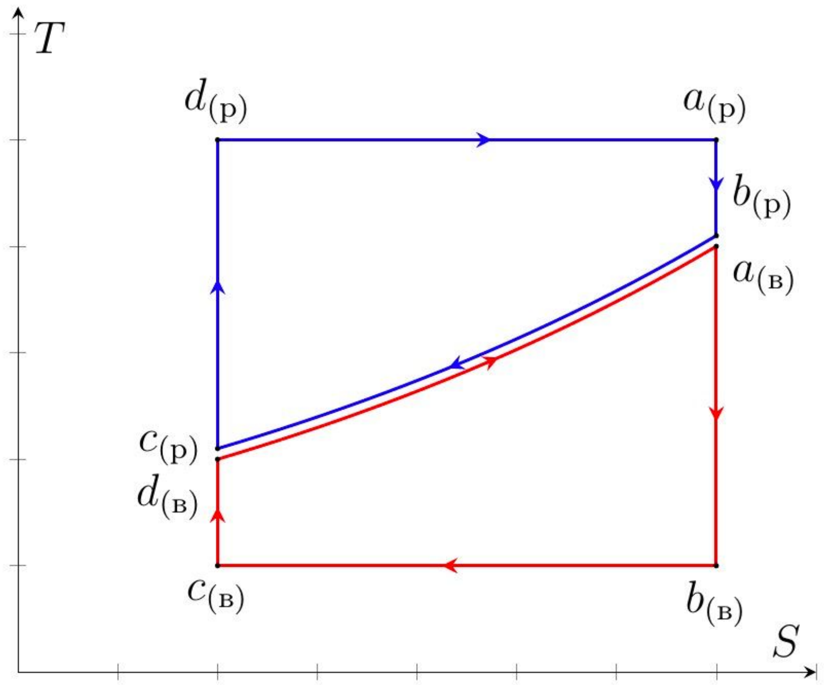
Since the amounts of heat received and given up by water and mercury vapor, respectively, in isobaric polytropic processes are the same at the same temperature changes, the heat capacities in isobaric processes are also the same:

$$C_{p(\mathrm{w})} = \frac{m_{(\mathrm{w})}\gamma_{(\mathrm{w})}R}{\mu_{(\mathrm{w})}(\gamma_{(\mathrm{w})} - 1)} = \frac{m_{(\mathrm{m})}\gamma_{(\mathrm{m})}R}{\mu_{(\mathrm{m})}(\gamma_{(\mathrm{m})} - 1)} = C_{p(\mathrm{m})}$$

from where:

Answer:

$$\frac{m_{(\mathrm{m})}}{m_{(\mathrm{w})}} = \frac{\mu_{(\mathrm{m})}}{\mu_{(\mathrm{w})}}\frac{\gamma_{(\mathrm{m})} - 1}{\gamma_{(\mathrm{m})}}\frac{\gamma_{(\mathrm{w})}}{\gamma_{(\mathrm{w})} - 1} = 17{,}81$$



B2 — 0.40 pts

Using graphs of the pressure of saturated vapors of mercury and water versus temperature, estimate the pressure values $p_{c(m)}$ and $p_{c(w)}$.

The required pressures are equal to the pressures of saturated vapors at temperatures t_4 for mercury and t_2 for water. From the graphs we find:

Answer:

$$p_{c(\mathrm{m})} = 6\ \mathrm{kPa}$$

B3 — 0.80 pts

Find the mercury and water cycle efficiencies $\eta_{(m)}$ and $\eta_{(w)}$ respectively.

Note: in all paragraphs below it is assumed that $T_i = t_i + T_0$, where $T_0 = 273$ K. The change in entropy in isothermal and isobaric processes is the same, therefore:

$$\Delta S = \int_{T_4}^{T_3} \frac{C_p dT}{T} = C_p \ln \frac{T_3}{T_4}$$

Amount heat, obtain mercury in isothermal process, equal:

$$Q_{+(\mathrm{m})} = T_1 \Delta S$$

Amount heat, given up water vapor in isothermal process, equal:

$$Q_{-(\mathrm{w})} = T_2 \Delta S$$

in isobaric process water pair obtain and mercury gives up heat Q_p, equal:

$$Q_p = C_p(T_3 - T_4)$$

therefore answer for $\eta_{(\mathrm{m})}$ and $\eta_{(\mathrm{w})}$ are as follows:

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therefore answer for $\eta_{(\mathrm{m})}$ and $\eta_{(\mathrm{w})}$ the following:

Answer:

$$\eta_{(\mathrm{m})} = 1 - \frac{T_3 - T_4}{T_1 \ln(T_3/T_4)} = 0{,}326$$

B4 — 0.80 pts

Find the efficiency η_{gv} of the binary gas-steam-water cycle. Compare it with $\eta_{(m)}$ and $\eta_{(w)}$.

Solution in TS coordinates.

Since in an isobaric process gases exchange heat with each other, heat is supplied from the outside to the system participating in the binary cycle only in the mercury isothermal process. Then the efficiency of the binary cycle is the same as that of the Carnot cycle, consisting of two isotherms and two adiabats with temperatures T_1 and T_2:

$$\eta_{\mathrm{Carnot}} = 1 - \frac{T_{\mathrm{c}}}{T_{\mathrm{h}}}$$

Finally:

Answer:

$$\eta_{\mathrm{gv}} = 1 - \frac{T_2}{T_1} = 0{,}657 > \eta_{(\mathrm{m})}, \eta_{(\mathrm{w})}$$

C1 — 0.40 pts

Find the temperatures of the system at points 2, 3, 4, 5 and 6.

Since the process 23 is isothermal, and the change in temperature in section 34 can be neglected by condition, the temperatures t_2, t_3 and t_4 are equal to the temperature of saturated steam at pressure $p_3 = 16$ kPa, i.e.:

The system temperatures at points 5 and 6 correspond to the temperature of saturated steam at pressure $p_4 = 2{,}8$ MPa, i.e.:

Answer:

$$t_2 = t_3 = t_4 = t_1' = 55°\mathrm{C}$$

C2 — 0.60 pts

Find the amount of heat Q_+ supplied to the system per cycle.

The figure shows a qualitative diagram of the cycle under consideration in TS coordinates.

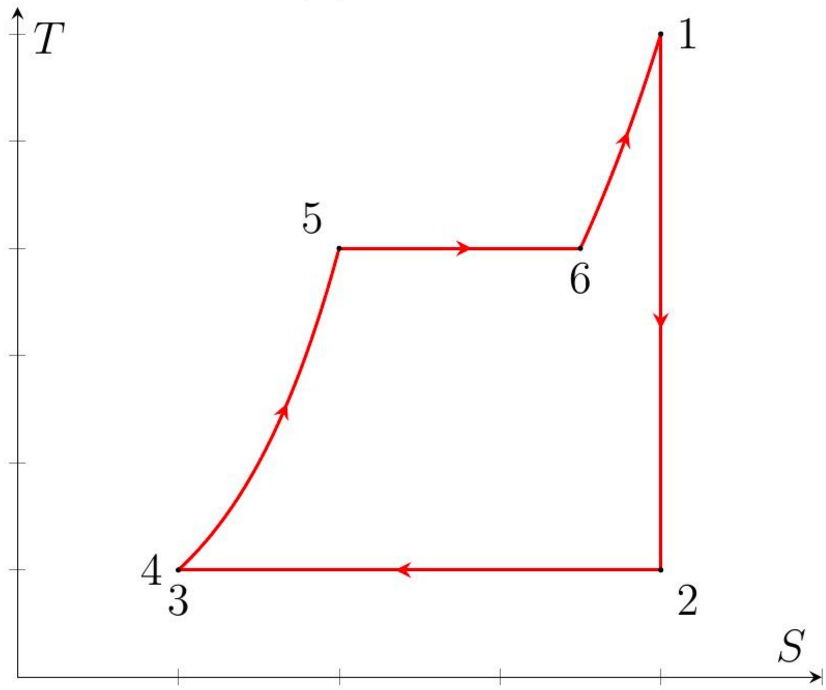
In section $4-5$ the liquid heats up, in section $5-6$ it evaporates, and section $6-1$ corresponds to the overheating of water vapor. Then for Q_+ we have:

$$Q_+ = cm(t_5 - t_4) + L(t_5)m + \frac{\gamma_{(\mathrm{w})} mR(t_1 - t_6)}{\mu_{(\mathrm{w})}(\gamma_{(\mathrm{w})} - 1)}$$

Since $t_6 > 180°\mathrm{C}$ - exponent/index adiabat water pair equal 9/7. Substituting numerical values, we find:

Answer:

$$Q_+ = 57{,}35\ \mathrm{MJ}$$



C3 0.80 pts

Find the amount of heat Q_- removed from the system per cycle.

The amount of heat given off per cycle in an isothermal process is equal to:

$$Q_- = T_2(S_2 - S_3)$$

Find change entropy other method, account for, that process $1-2$ and $3-4$ be adiabatic:

$$S_2 - S_3 = S_5 - S_4 + S_6 - S_5 + S_1 - S_6 = m\left(c\ln\frac{T_5}{T_4} + \frac{L(t_5)}{T_5} + \frac{\gamma_{(\mathrm{w})}R}{\mu_{(\mathrm{w})}(\gamma_{(\mathrm{w})} - 1)}\ln\frac{T_1}{T_6}\right)$$

Thus:

$$Q_- = mT_2\left(c\ln\frac{T_5}{T_4} + \frac{L(t_5)}{T_5} + \frac{\gamma_{(\mathrm{w})}R}{\mu_{(\mathrm{w})}(\gamma_{(\mathrm{w})} - 1)}\ln\frac{T_1}{T_6}\right)$$

Substituting the numerical values, we find:

Answer:

$$Q_- = 36{,}83\ \mathrm{MJ}$$

C4 0.20 pts

Find the gas work A per cycle and its efficiency η_2.

Let's use the law of conservation of energy:

$$A = Q_+ - Q_-$$

from where:

By definition:

$$\eta_2 = \frac{A}{Q_+} = 1 - \frac{Q_-}{Q_+}$$

from:

Answer:

$$A = 20{,}52 \text{ MJ}$$

D1 0.20 pts

Find the temperatures at points b_{in}, c_{in}, d_{in}, e_{in}, f_{in} of the water cycle.

The temperatures at points $b_{(\text{in})}$, $c_{(\text{in})}$ and $d_{(\text{in})}$ are the same and equal to the temperature of saturated water vapor at pressure $p_{c(\text{in})}$:

The temperatures at points $e_{(\text{in})}$ and $f_{(\text{in})}$ are the same and equal to the temperature of saturated water vapor at pressure $p_{e(\text{in})}$:

Answer:

$$t_{b(\text{in})} = t_{c(\text{in})} = t_{d(\text{in})} = t_1 = 96^\circ\text{C}$$

D2 1.00 pts

Find the temperature at point a_{in} of the water cycle.

The figure shows qualitative diagrams of the considered cycles in TS coordinates.

Since the processes $a_{\text{in}} - b_{\text{in}}$ and $c_{\text{in}} - d_{\text{in}}$ are adiabatic, the change in entropy in sections $d_{\text{in}} - e_{\text{in}} - f_{\text{in}} - a_{\text{in}}$ is equal to the change in entropy in section $c_{\text{in}} - b_{\text{in}}$:

$$S_{b(\text{w})} - S_{c(\text{w})} = S_{e(\text{w})} - S_{d(\text{w})} + S_{f(\text{w})} - S_{e(\text{w})} + S_{a(\text{w})} - S_{f(\text{w})}$$

For change entropy water on section $d_{\text{in}} - e_{\text{in}} - f_{\text{in}} - a_{\text{in}}$ have:

$$S_{a(\text{w})} - S_{d(\text{w})} = m_{\text{in}} \left(c_{\text{in}} \ln \frac{T_2}{T_1} + \frac{L_{\text{in}}(T_2)}{T_2} + \frac{\gamma_{\text{in}} R}{\mu_{\text{in}}(\gamma_{\text{in}} - 1)} \ln \frac{T_{a(\text{w})}}{T_2} \right)$$

For change entropy on section $c_{\text{in}} - b_{\text{in}}$ have:

$$S_{b(\text{w})} - S_{c(\text{w})} = \frac{m_{\text{in}} L_{\text{in}}(T_1)}{T_1} + \frac{A_{\text{pair}}}{T_1}$$

A_{par} – work done by steam during expansion. Let's find it:

$$A_{\text{pair}} = \int\limits_{V_0}^{V_{\text{to}}} p\,dV = \nu RT \int\limits_{V_0}^{V_{\text{to}}} \frac{dV}{V} = \nu RT \ln \frac{V_{\text{to}}}{V_0}.$$

Since steam expands isothermally:

$$\frac{V_{\text{to}}}{V_0} = \frac{p_0}{p_{\text{to}}} = \frac{p_{c(\text{w})}}{p_{b(\text{w})}}.$$

We obtain an equation containing T_a:

$$\frac{L_{\text{in}}(T_1)}{T_1} + \frac{R}{\mu_{\text{in}}} \ln \frac{p_{c(\text{w})}}{p_{b(\text{w})}} = c_{\text{in}} \ln \frac{T_2}{T_1} + \frac{L_{\text{in}}(T_2)}{T_2} + \frac{\gamma_{\text{in}} R}{\mu_{\text{in}}(\gamma_{\text{in}} - 1)} \ln \frac{T_{a(\text{w})}}{T_2}.$$

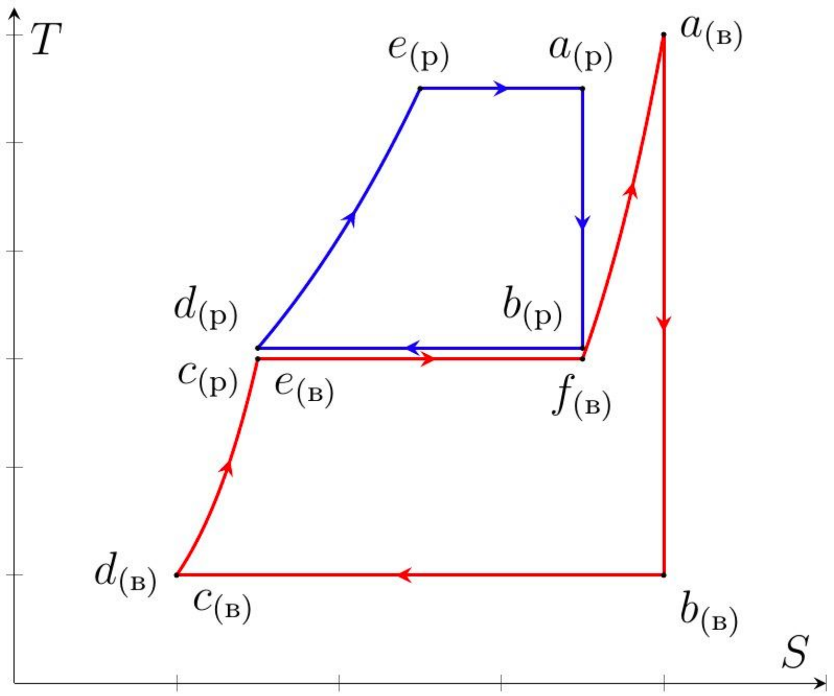
Substituting numerical values for $L_{\text{in}}(T_1)$ and $L_{\text{in}}(T_2)$:

$$L_{\text{in}}(T_1) = 2{,}27 \text{ MJ} \qquad L_{\text{in}}(T_2) = 1{,}98 \text{ MJ}$$

Finally we get:

Answer:

$$t_{a(\text{w})} \approx 677°\text{C}$$



D3 — 1.30 pts

Find the mercury pressure at section $d_m - e_m - a_m$.

Let us denote the temperature at point e_{m} as T_4. Then for the change in entropy in section $d_{\text{m}} - e_{\text{m}} - a_{\text{m}}$ we have:

$$S_{a(\text{m})} - S_{d(\text{m})} = m_{\text{m}} \left(c_{\text{m}} \ln \frac{T_4}{T_2} + \frac{L_{\text{m}}(T_4)}{T_4} \right)$$

On the other hand, this same change entropy equal change entropy on section $c_{\text{m}} - b_{\text{m}}$, which, in its own turn, equal change entropy water on section $e_{\text{in}} - f_{\text{in}}$:

$$S_{f(\text{w})} - S_{f(\text{w})} = \frac{m_{\text{in}} L_{\text{in}}(T_2)}{T_2}$$

Equate, obtain dependence $L_{\text{m}}(T_4)$:

$$L_{\text{m}}(T_4) = \frac{m_{\text{in}} L_{\text{in}}(T_2) T_4}{m_{\text{m}} T_2} - c_{\text{m}} T_4 \ln \frac{T_2}{T_4}$$

Construct given function on boundary dependence $L_{\text{m}}(t)$ and find solution:

$$t_4 \approx 610°\text{C}$$

Immediately determinable $L_{\text{m}}(t_4)$:

$$L_{\text{m}}(t_4) \approx 288 \text{ kJ/kg}$$

The pressure of mercury in the desired area is equal to the pressure of saturated vapor of mercury at a given temperature. Finally we have:

Answer:

$$p_{e(\mathrm{m})} \approx 2{,}6\ \mathrm{MPa}$$

D4 1.00 pts

Find the efficiency of the binary Rankine cycle with mercury and water η_3.

Mercury in section $b_\mathrm{m} - c_\mathrm{m}$ exchanges heat with water in section $e_\mathrm{in} - f_\mathrm{in}$, so heat is supplied from the outside to mercury in section $d_\mathrm{m} - e_\mathrm{m} - a_\mathrm{m}$, as well as to water in sections $d_\mathrm{in} - e_\mathrm{in}$ and $f_\mathrm{in} - a_\mathrm{in}$. Then for the amount of heat supplied per unit mass of water Q_{+3}/m_in we obtain:

$$\frac{Q_{+3}}{m_\mathrm{in}} = \left(c_\mathrm{in}(t_2 - t_1) + \frac{\gamma_\mathrm{in} R(t_3 - t_2)}{\mu_\mathrm{in}(\gamma_\mathrm{in} - 1)}\right) + \frac{m_\mathrm{m}}{m_\mathrm{in}}(c_\mathrm{m}(t_4 - t_2) + L_\mathrm{m}(t_4)) \approx 4{,}976\ \frac{\mathrm{MJ}}{\mathrm{kg}}$$

Heat in system is removed only from water on section $b_\mathrm{in} - c_\mathrm{in}$, therefore for reduced amount heat on unitat mass water Q_{-3}/m_in have:

$$\frac{Q_{-3}}{m_\mathrm{in}} = \frac{T_1(S_{b(\mathrm{w})} - S_{c(\mathrm{w})})}{m_\mathrm{in}}$$

From result part D2:

$$\frac{S_\mathrm{bin} - S_\mathrm{cin}}{m_\mathrm{in}} = \left(c_\mathrm{in} \ln\frac{T_2}{T_1} + \frac{L_\mathrm{in}(T_2)}{T_2} + \frac{\gamma_\mathrm{in} R}{\mu_\mathrm{in}(\gamma_\mathrm{in} - 1)} \ln\frac{T_{a(\mathrm{w})}}{T_2}\right)$$

From which:

$$\frac{Q_{-3}}{m_\mathrm{in}} = T_1\left(c_\mathrm{in} \ln\frac{T_2}{T_1} + \frac{L_\mathrm{in}(T_2)}{T_2} + \frac{\gamma_\mathrm{in} R}{\mu_\mathrm{in}(\gamma_\mathrm{in} - 1)} \ln\frac{T_{a(\mathrm{w})}}{T_2}\right) \approx 2{,}484\ \frac{\mathrm{MJ}}{\mathrm{kg}}$$

Final:

Answer:

$$\eta_3 \approx 0{,}50$$